13 A very hot mug of coffee is near a large unheated swimming pool holding 2.0×10^6 kg of water.

Which statement is correct?

A If the mug of coffee is tipped in the pool, the coffee will lose much more internal energy than the water in the pool gains.

B The internal energy of a substance depends only on its temperature.

C The internal energy of the water in the pool is lower than the internal energy of the mug of coffee.

D When the temperature of the mug of coffee falls, its internal energy will decrease.